

A three-way distinction of negation interpretation in Czech

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The present article proposes that negation in Czech occupies three distinct LF positions: outer, medial and inner. They differ with respect to their licensing conditions, meaning imports, overt syntactic marking as well as in their distribution across clause types. I identify the three readings of negation in the material of Czech polar questions, which allows for all of them. While inner negation is akin to standard sentential negation, medial and outer operate on the pragmatic level. Both medial and inner negation interact with a newly proposed modal operator deriving evidential bias in PQs. Building on Repp (2013) and Šimík (to appear), I analyze outer negation as the *FALSUM* operator. I propose that *FALSUM* in Czech PQs is obligatorily focused. The account has further implications for the encoding of information structure in Czech negative polar questions.

Keywords: negation, Czech, polar questions, bias, polarity focus, information structure

1 Introduction

It is traditionally assumed that negation in English polar questions (= PQs) might overtly occur *HIGH* (*Doesn't John smoke?*) or *LOW* (*Does John not smoke?* / *John doesn't smoke?*). Apart from these, two readings of negation have been distinguished: *INNER* and *OUTER* (Romero & Han (2004), building on Ladd (1981)). Ladd's 1981 original terms are *OUTSIDE* and *INSIDE* negation, where a PQ with outside negation questions the prejacent p , while a PQ with inside negation questions $\neg p$. In the present article, I follow the approach of Romero & Han (2004) or Holmberg (2013), who distinguish *INNER* and *OUTER* negation on LF, where inner negation merges in the proposition, while outer negation merges outside of the proposition. For Romero & Han (2004), the two negation readings denote that the speaker either believes $\neg p$ (*INNER*) or p (*OUTER*) when asking the question.

The syntactic positions have been noticed to be associated with the respective interpretations in English. For Romero & Han (2004), low negation is interpreted as inner, but high negation is ambiguous between the inner and outer readings. According to AnderBois (2019) and Goodhue (2022), low negation receives the inner reading, while high negation receives outer. These form-meaning mappings have been elaborated by Holmberg (2013), who argues for a third reading of negation in English (calling it *MIDDLE*), which maps onto either high or low negation.

For Czech PQs, Staňková & Šimík (2025) showed experimentally that high negation receives only the outer reading, as evidenced by the unavailability of the negative concord item *žádné*, see (1a), while low negation is ambiguous, as evidenced by the availability of both

negative concord items ($\approx$ inner negation) and positive polarity items (without outscoping negation; $\approx$ outer negation) like *nějaké*, see (1b).¹

- (1) a. Nezasadila tam Dana {nějaké / *žádné} květiny?
 NEG.planted there Dana DET.PPI DET.NCI flowers
 ‘Didn’t Dana plant there some flowers?’
 b. Dana tam nezasadila {nějaké / žádné} květiny?
 Dana there NEG.planted DET.PPI DET.NCI flowers
 ‘Dana didn’t plant there some/any flowers?’

I aim to clarify the ambiguity of low negation by looking more closely at the possible interpretations negation can receive in Czech PQs and other grammatical environments. In the present article, I pose the following research questions: (i) what are the positions of negation interpretation in Czech?; and (ii) how is the interpretation of negation linked to the information structure of an utterance? Based on new data and observations, I propose that the picture is richer than previously expected, and that negation in Czech might occupy three distinct LF positions. Based on their relative position, I call them INNER, MEDIAL and OUTER. Their proposed positions are shown in (2). It will be argued that while inner negation operates on the level of propositions, medial and outer operate on the pragmatic level.

- (2) [CP ... [PolP *ne-* [ModP *ne-* [TP *ne-*]]]]
 OUTER MEDIAL INNER

I will show that the three readings of negation in Czech differ in their: (i) compatibility with polarity items, (ii) ability to license certain particles (*náhodou* ‘by chance’, *ještě* ‘yet’ and *fakt* ‘really’), (iii) interaction with other components, e.g. the modal component of evidential bias, and (iv) syntactic and prosodic marking (word order and pitch accenting). The account has further implications as to how information structure (esp. focus) is encoded in Czech negative PQs.

In Section 2 the three readings of negation are distinguished by their compatibility with positive polarity items, negative concord items and the above-mentioned particles. Based on these observations, in Section 3 I propose that they occupy distinct LF positions and have different meaning imports. The ways focus is encoded in negative PQs is presented in Section 4. Section 5 concludes.

2 Czech negation and licensing

Czech is usually claimed to be a language with very flexible word order (Siewierska & Uhliřová 1998). Movement of the constituents is not completely free though, as it is conditioned by several factors. One of them is the fixed position of clitics (Wackernagel (1892)/2020), and there are also word order configurations induced by certain information structural categories, such as givenness (Biskup 2006) or focus. The canonical declarative word order in Czech assertions is SVO, whereas the default interrogative word order in Czech PQs is VSO (Šimík to appear).

Negation in Czech is marked by the prefix *ne-*, which is attached to the inflected verb. This prefix is not independent, so wherever the verb moves in the structure, the prefix moves with it. In what follows I present empirical issues which I explain by proposing that negation in

¹In the examples, I gloss positive polarity items as PPI and negative concord items as NCI.

Czech can receive three different interpretations – I call them *INNER*, *MEDIAL*, and *OUTER* and distinguish them on the LF level. First, I discuss the compatibility of negation with polarity items.

2.1 Compatibility with polarity items

It is commonly assumed that sentential negation is contributed by the semantic operator $\neg$. This operator has been previously claimed to reside in a dedicated phrase called *NegP* in Czech sentences. *NegP* is located between *TP* and *AspP* in the extended verbal projection (Kosta 2001; Biskup 2017); see (3). An example of an assertion with $\neg$ with its denotation is in (4).

(3) $[_{CP} C [_{TP} T [_{NegP} Neg [_{AspP} Asp [_{vP} v [_{VP} V]]]]]]$

(4) $\llbracket \text{Eva didn't sleep} \rrbracket = \llbracket Neg [\text{Eva slept}] \rrbracket = \lambda w. \neg \text{SLEPT}(\text{EVA}, w)$

According to Zeijlstra (2004), Czech is a strict negative concord language. This means that negative concord items (= NCIs) such as *žádný* ‘no.DET.NCI’ or *nic* ‘nothing.NCI’ and the negative verb, all inherently non-negative, need to be licensed by $\neg$ to derive a felicitous utterance. It is assumed that there is an Agree relation between $\neg$ carrying [iNeg] and the [uNeg] probes. The local domain of licensing NCIs is inside a clause, as they need to be in the scope of $\neg$. PPIs (such as *nějaký* ‘some.DET.PPI’), on the contrary, must outscope $\neg$ in order to be interpreted. This is shown in (5), where both replies are grammatical, but have different meanings because of the scope relations between the negation and the polarity item. Interpreting PPIs in the scope of $\neg$ leads to infelicity in assertions.

- (5) a. Eva si nepustila žádný podcast. $\neg > \exists_{\text{NCI}}$
 Eva REFL NEG.played DET.NCI podcast
 ‘Eva didn’t listen to any podcast.’
 b. Eva si nepustila nějaký podcast. $\exists_{\text{PPI}} > \neg / * \neg > \exists_{\text{PPI}}$
 Eva REFL NEG.played DET.PPI podcast
 ‘There was some podcast that Eva didn’t not listen to.’
 # ‘Eva didn’t listen to some podcast.’

In PQs, NCIs are licensed only when the negation is low, as illustrated in (6). High negation does not allow NCIs in its scope; it anti-licenses NCIs, which has been experimentally tested and confirmed for Czech by Staňková & Šimík (2025). This indicates that only low negation is able to receive the sentential negation reading, whose scope is limited to the proposition. In PQs, I call this negation *INNER*.

- (6) a. Petr si nekoupil žádný časopis?
 Petr REFL NEG.bought DET.NCI magazine
 Petr didn’t buy any magazine?
 b. *Nekoupil si Petr žádný časopis?
 NEG.bought REFL Petr DET.NCI magazine
 Intended: ‘Didn’t Petr buy a magazine?’

Similarly to assertions, PQs can also involve wide-scoping PPIs in Czech. Unlike in assertions, however, PQs are also amenable to a reading where the PPI does not outscope negation, suggesting that low negation in PQs can be interpreted as non-inner (non-propositional). This is shown in (7) with both low and high negation. I take these facts to be an indication of the negation(s) in (7) having wider LF scope than inner negation. In Section 3 I propose that (7a)

and (7b) are represented higher than inner negation on LF, and I call them MEDIAL and OUTER respectively.²

- (7) *Scenario*: Eva and her colleagues from the office ordered take-out food for lunch. The delivery guy brings only pizzas. One of the colleagues asks:
- a. Eva si neobjednala nějaký burger? NEG > $\exists_{\text{PPI}}$
Eva REFL NEG.ordered DET.PPI burger
‘Eva didn’t order some burger?’
 - b. Neobjednala si Eva nějaký burger? NEG > $\exists_{\text{PPI}}$
NEG.ordered REFL Eva DET.PPI burger
‘Didn’t Eva order some burger?’

Summing up this section, I showed that Czech PQs display two word order variants: declarative and interrogative. Negation in a declarative PQ might be outscoped by a PPI, in which case I call it inner, but it also might scope over a PPI, in which case I call it medial. Inner negation licenses NCIs by an Agree relation. Medial negation is non-propositional and so is predicted not to license NCIs. Negation in an interrogative PQ always scopes over a PPI, but cannot license an NCI; I call it outer negation. My proposal is that inner, medial, and outer negations differ at LF, and that their interaction with polarity items as well as word order indicate this. Another reason for distinguishing them is their (in)compatibility with certain particles – I use *náhodou* (‘by chance’), *ještě* (‘yet’) and *fakt/určitě/opravdu* (‘really’) as diagnostics of the three LF negations.

2.2 Licensing of particles

2.2.1 *Náhodou*

The particle *náhodou* (‘by chance’) occurs exclusively in negative PQs (Štícha 1984; Daneš et al. 1987; Chodounská et al. 2025), where its presence is conditioned by the so-called **SPEAKER BIAS** (although e.g. Sudo 2013 calls this notion **EMPIRICAL BIAS**).³ A PQ carries speaker bias if the speaker implies that they hold an epistemic attitude to one of the possible answers, for instance (and typically) if the speaker believes either p or $\neg p$. Speaker bias is primarily based on the speaker’s prior attitude, and it is contrasted with **EVIDENTIAL BIAS**, which is based on the current contextual evidence (more on that in Section 3.1).

Speaker bias is present in Czech negative PQs, when the negation is high (in the initial position), which receives exclusively the outer reading (Staňková & Šimík 2025); shown in (8). For Czech PQs with high negation, Šimík (to appear) proposes that the negative marker on the verb contributes epistemic possibility at a non-at-issue level in the form of an existential epistemic modal. That is, from (8) it is inferrable that the speaker thinks that the prejacent p (i.e., that Eva borrowed an encyclopedia) is *possible*, but they do not necessarily *believe* that p

²An anonymous reviewer points out that the outer vs. inner distinction is often understood via epistemic bias: outer negation implies that the speaker believes p , and inner negation does not (or, it implies that the speaker believes $\neg p$). In this paper I take a cautious distance from this stance and instead emphasize that Czech appears to necessitate three types of semantic/LF negations in PQs. That said, the three negations do correlate with different bias patterns in Czech (as I show below), but these are not necessarily of the same kind found in English. See, e.g., Šimík (to appear), for an argument that high negation in English and Czech have comparable semantics (justifying the use of the term “outer” negation), but crucially not the same semantics, calling for a more cautious treatment of negation in PQs.

³*Náhodou* also works as an adverbial in Czech meaning ‘accidentally’. I am concerned only with its particle usage in this article.

(this inference arises due to high negation alone, not due to the particle *náhodou*). *Náhodou* is hypothesized to modify the ordering source of the existential epistemic modal found in high negation PQs by including less stereotypical worlds in the modal's domain of quantification. Its occurrence depends on the presence of outer negation and, consequently, on the weak (existential) speaker bias.

- (8) *Scenario*: Eva is doing research for a school presentation. Yesterday she went to the school library. The speaker knows all this from before and asks:

Nepůjčila si tam Eva náhodou {nějakou / #žádnou} encyklopedii? OUTER
 NEG.borrowed REFL there Eva NÁHODOU DET.PPI DET.NCI encyclopedia
 'Didn't Eva borrow some encyclopedia, by any chance?'

The examples in (9a) and (9b) show that low negation is not compatible with *náhodou* (unless the utterance features a contrastive topic, see example ((24)-B₂) and the discussion in Section 3.2).⁴ If *náhodou* depends on the presence of weak speaker bias in the PQ, then these data suggest that low negation PQs do not necessarily signal speaker bias (that their interpretation does not depend on it). I come back to the question of bias that the individual negation readings signal in Section 3.1 and Section 3.2. In any way, *náhodou* seems to be a marker of the specific reading that high negation is associated with – outer negation.

- (9) *Scenario*: Eva is doing research for a school presentation. Yesterday she went to the school library, but left with her backpack empty. The speaker found out about her having left with an empty backpack and asks:

- a. Eva si (#náhodou) nepůjčila nějakou encyklopedii? MEDIAL
 Eva REFL NÁHODOU NEG.borrowed DET.PPI encyclopedia
 'Eva didn't borrow some encyclopedia?'
- b. Eva si (#náhodou) nepůjčila žádnou encyklopedii? INNER
 Eva REFL NÁHODOU NEG.borrowed DET.NCI encyclopedia
 'Eva didn't borrow any encyclopedia?'

2.2.2 *Ještě*

The felicity of a sentence with *ještě* 'still'/'yet' depends on the properties of the predicate. In the case of atelic predicates, *ještě* can, but does not have to cooccur with negation, as shown in (10a) and (10b). When the predicate is telic, *ještě* requires negation, as shown in (10c) and (10d).^{5,6}

⁴Here I focus mostly on matrix PQs, but *náhodou* and outer negation are likewise possible in embedded PQs:

- (i) Zeptala se mě, jestli si Petr náhodou nekoupil nějaké auto.
 asked REFL me whether REFL Petr NÁHODOU NEG.bought DET.PPI car
 'She asked me whether Petr has bought some car.'

⁵I leave aside the issue of whether *ještě* in combination with telic predicates is a narrow-scoping negative polarity item, or whether it is incompatible with telic predicates in the first place and negation yields an atelic one.

⁶*Ještě* also has an additive meaning (irrelevant for current purposes), which can be translated as 'again' or 'once more'. On that reading, example (10c) is grammatical and can be translated as 'Petr fell asleep again/once more.' For a similar reading of the German *noch* 'still', see Grubic & Wierzba (2021).

- (10) a. Petr ještě spí. atelic predicate
 Petr JEŠTĚ sleeps
 ‘It is still the case that Petr is sleeping.’
 b. Petr ještě nespí. negative atelic predicate
 Petr JEŠTĚ NEG.sleeps
 ‘It is still the case that Petr isn’t sleeping.’ / ‘Petr isn’t sleeping yet.’
 c. *Petr ještě usnul. telic predicate
 Petr JEŠTĚ fell-asleep
 Intended: ‘It is still the case that Petr has fallen asleep.’
 d. Petr ještě neusnul. negative telic predicate
 Petr JEŠTĚ NEG.fell-asleep
 ‘It is still the case that Petr hasn’t fallen asleep.’ / ‘Petr hasn’t fallen asleep yet.’

The behavior of *ještě* in PQs reveals that, when combined with telic predicates, it is sensitive to inner negation, on a par with negative concord items. Example (11a) is therefore ungrammatical (on the relevant reading), on a par with (10c). In negative PQs, it is felicitous with inner negation, diagnosed either by the presence of an NCI, as in (11b), or a wide-scoping PPI, as in (11c). *Ještě* is not felicitous in a PQ where the negation scopes over a PPI, which is shown in (11d) and (11e), for medial and outer negation, respectively. The behavior of *ještě* in PQs thus clearly distinguishes between inner negation on the one hand, and medial/outer on the other.⁷

- (11) a. *Petr ještě složil zkoušku?
 Petr JEŠTĚ passed exam
 Intended: ‘Is it still the case that Petr has passed an exam?’
 b. Petr ještě nesložil žádnou zkoušku? $\neg > \exists_{\text{NCI}}$
 Petr JEŠTĚ NEG.passed DET.NCI exam
 ‘Is it still the case that Petr hasn’t passed any exam?’
 c. Petr ještě nesložil nějakou zkoušku? $\exists_{\text{PPI}} > \neg$
 Petr JEŠTĚ NEG.passed DET.PPI exam
 ‘Is it still the case that Petr hasn’t passed a certain exam?’
 d. *Petr ještě nesložil nějakou zkoušku? MEDIAL NEG
 Petr JEŠTĚ NEG.passed DET.PPI exam
 Intended: ‘Is it still the case that Petr hasn’t passed any exam?’
 e. *Nesložil Petr ještě nějakou zkoušku? OUTER NEG
 NEG.passed Petr JEŠTĚ DET.PPI exam
 Intended: ‘Isn’t it that it is still the case that Petr has passed an exam?’

2.2.3 Fakt

Finally, I discuss the compatibility of PQ negation with the particle *fakt* ‘really’. The set of empirical data showing its felicity alongside negation is in (12).⁸

- (12) a. Eva fakt neviděla žádný Taratinův film? $\text{FAKT} > \neg > \exists_{\text{NCI}}$
 Eva FAKT NEG.saw DET.NCI Tarantino’s film
 ‘Eva really hasn’t seen any film by Tarantino?’

⁷I point out for completeness that (11a) and (11e) are grammatical on the irrelevant additive reading; see footnote 6.

⁸Example (12d) is grammatical on a reading where *fakt* contributes a meaning close to ‘after all’: ‘Hasn’t Eva seen a film by Tarantino, after all?’

LF negation	NE- > PPI	NCI	<i>náhodou</i>	<i>ještě</i>	<i>fakt</i>
outer	✓	✗	✓	✗	✗
medial	✓	✗	✗	✗	✓
inner	✗	✓	✗	✓	✓

Table 1: Negation readings in Czech and their compatibility with PIs and particles

- b. Eva fakt neviděla nějaký Tarantinův film? FAKT > $\exists_{\text{PPI}}$ > $\neg$
 Eva FAKT NEG.saw DET.PPI Tarantino's film
 'There is really some film by Tarantino that Eva hasn't seen?'
- c. Eva fakt neviděla nějaký Tarantinův film? FAKT > MEDIAL NEG
 Eva FAKT NEG.saw DET.PPI Tarantino's film
 'Is it really the case that Eva hasn't seen a film by Tarantino?'
- d. *Neviděla Eva fakt nějaký Tarantinův film? *FAKT > OUTER NEG
 NEG.saw Eva FAKT DET.PPI Tarantino's film
 Intended: 'Is it really not the case that Eva has seen a film by Tarantino?'

From the present data it seems that only low negation licenses *fakt* (in (12a), (12b) and (12c)), whereas high negation repels it (in (12d)). In other words, when negation in a PQ is interpreted as inner or medial, *fakt* is allowed, but when the outer reading is present, *fakt* (at least on its canonical reading) is not allowed.

I leave the semantic contribution of *fakt* (and related particles like *opravdu* 'really/truly' and *určitě* 'surely') for future research. However, it should be noted that *fakt* bears similarity to the Russian question particle *razve* and to its English counterpart *really*. *Razve* seems to pattern with *fakt* in Czech low negation PQs, as it is compatible with NPIs as well as PPIs in its scope (Repp & Geist 2025, p. 9). *Really* has been extensively studied by others before. Romero & Han (2004) propose that *really* (just like high negation and polarity focus) maps to the conversational epistemic operator VERUM. In positive PQs, *really* conveys the implicature that the speaker believed or expected that $\neg p$ was true (i.e., speaker bias for $\neg p$). Goodhue (2022) derives the same speaker bias for *really* via a pragmatic route. While the properties of *fakt* and its relation to the better-studied particles are yet to be explored, what is important for the present purposes is that it distinguishes between outer negation on the one hand and medial and inner on the other.

3 Proposal

On the material of Czech PQs I showed that negation might set several licensing conditions on polarity items and the particles *náhodou*, *ještě* and *fakt*. I take the PQs' compatibility with these elements and their combination to be indicative of the three distinct types of negation interpretation in Czech PQs. These are summarized in 1.

My concrete proposal for the mutual positions of outer, medial and inner negation in Czech PQs in LF syntax is in (13). The placement of negations in the LF structure reflects their respective scope. Inner negation has the narrowest scope and licenses NCIs. Medial negation scopes outside of the proposition and so allows for PPIs to be interpreted under it. Both the inner and medial readings map onto low negation, but they differ in the interaction with the evidential bias modal (more on that in the upcoming Section 3.1). Outer negation has the widest scope and allows for PPIs under it. It maps onto high negation and its interpretation

relies on speaker bias (more on that in Section 3.2).

- (13) [CP ... [PolP *ne-* [ModP *ne-* [TP *ne-*]]]]
 OUTER MEDIAL INNER

In the following sections I focus on each of these three readings separately, describing their semantic/pragmatic import in PQs. I will show that each of them has unique contextual requirements and thus leads to a different bias structure. Importantly, these facts will reflect the proposed LF structure.

3.1 Medial and inner negation – relation to contextual evidence

The experimental results of Staňková & Šimík (2025p. 9) showed that Czech low negation PQs are more natural in contexts (or situations) which contain some evidence against *p* than in contexts which do not contain such evidence. The examples in (14a) and (14b) with the inner and medial negation reading illustrate the compatibility of low negation with contextual evidence against *p*. These PQs would, however, not sound natural in a situation where the students are about to write a test, and the speaker either cannot see Petr and what he is doing (lack of contextual evidence) or sees that Petr actually brought a pen (contextual evidence for *p*). Such PQs as in (14a) and (14b) are said to carry evidential bias.⁹

- (14) *Scenario*: Petr and his classmates are about to write a test at school. Everybody is ready, but Petr is asking the classmates sitting next to him to borrow him a pen. The speaker of the PQ sits across the classroom, sees this and asks:
- a. Petr si nevzal žádnou tužku? INNER
 Petr REFL NEG.brought DET.NCI pen
 ‘Petr didn’t bring any pen?’
- b. Petr si nevzal nějakou tužku? MEDIAL
 Petr REFL NEG.brought DET.PPI pen
 ‘Petr didn’t bring some pen?’

Evidential bias is an expectation about the answer to the current PQ on the side of the speaker. It is based on contextual/situational evidence which is (i) mutually available to both speakers, (ii) relatively new and (iii) compelling (Büring & Gunlogson 2000). Similarly to speaker bias, I take it to be part of non-at-issue meaning in PQs, although qualitatively they are distinct. I analyze evidential bias as presuppositional content and represent it as a modal operator $\Box_{ev}$ which quantifies over a modal base and an ordering source in the sense of Kratzer (1991). In the case of $\Box_{ev}$, the ordering source is a stereotypical one and the modal base consists of the context set (Stalnaker 2002). Since evidentially biased PQs require a specific kind of contextual evidence, I assume the modal force to be that of necessity.

In the previous sections I have claimed that low negation might receive two different readings – inner and medial – and that the former occupies a lower LF position than the latter. Now I will show that their LF positions reflect the strength of evidential bias such PQs carry: inner negation PQs carry strong evidential bias, whereas medial negation PQs carry weak evidential bias. An important piece of empirical evidence supporting the claim is presented in (15). In a context (or situation) which presents very clear contextual evidence for $\neg p$, only

⁹Czech positive PQs can also carry evidential bias. What they have in common with low negation PQs is the declarative word order.

an inner negation PQ is felicitous; see (15a) which actually fulfills the role of a rhetorical PQ, since the issue whether p or $\neg p$ holds is already resolved for $\neg p$ in the context. While medial negation PQs are only compatible with contextual evidence against p , as in (14b), inner negation PQs compatible with contextual evidence against p (14a) as well as for $\neg p$. This suggests that inner negation PQs carry strong negative evidential bias, but medial negation PQs carry weak negative evidential bias, as they cannot be used rhetorically.

(15) *Scenario*: Petr was supposed to buy a birthday cake for his friend's birthday party, but he forgot about it and didn't buy any. The speaker comes to the party, sees no cake (finds out about Petr having forgotten) and asks:

- | | | |
|----|---|--------|
| a. | Petr nekoupil žádný dort?
Petr NEG.bought DET.NCI cake
'Petr didn't buy any cake?' | INNER |
| b. | #Petr nekoupil nějaký dort?
Petr NEG.bought DET.PPI cake
'Petr didn't buy some cake?' | MEDIAL |

In what follows, I account for this fact by the negations' interaction with the modal $\square_{ev}$. Inner negation scopes under $\square_{ev}$ and stays at the level of propositional meaning. On the contrary, medial negation scopes above $\square_{ev}$ and becomes a part of the evidential bias, i.e., presuppositional meaning.

Inner negation in PQs signals evidential bias for $\neg p$. When a PQ is evidentially biased for $\neg p$, there has to be some compelling evidence for $\neg p$ in the context or situation – see (14a), where asking for a pen counts as evidence for $\neg p$ (= that Petr did not bring a pen). I propose that in biased PQs with inner negation like (14a), the negation scopes under $\square_{ev}$, that is at the propositional level. The speaker conveys that based on the evidence present in the context, it must be that $\neg p$ holds. I capture this presuppositional content of evidential bias in (16):

- (16) $\llbracket \square_{ev}(\text{NEG } p) \rrbracket$ is defined iff $\forall w' \in \text{BEST}_{g(w)}(\bigcap f(w)) : \neg p(w') = 1$
If defined, then $\llbracket \square_{ev}(\text{NEG } p) \rrbracket = \lambda w[\neg p(w)]$

Informally: based on the context set, it must be that $\neg p$ holds ($\square_{ev} > \neg$)

For medial negation I propose that it scopes over $\square_{ev}$. The formal representation of medial negation and $\square_{ev}$ is in (17). Medial negation scopes high and is a part of the presuppositional meaning of a PQ. The fact that this kind of modality can occur in the scope of negation suggests that its nature is rather that of root or circumstantial modality rather than epistemic.¹⁰

- (17) $\llbracket \text{NEG } \square_{ev}(p) \rrbracket$ is defined iff $\neg \forall w' \in \text{BEST}_{g(w)}(\bigcap f(w)) : p(w') = 1$
If defined, $\llbracket \text{NEG } \square_{ev}(p) \rrbracket = \lambda w[p(w)]$

Informally: based on the context, it does not have to be that p holds ($\text{NEG} > \square_{ev}$)

Both inner and medial negation share the same overt position (low) and need to be used in contexts biased against p , although the bias is stronger for inner negation than in medial negation. Importantly, neither of them is tied to a specific value of speaker bias. This is in accord with previous claims about negative declarative PQs in English: Gunlogson (2002)

¹⁰Semantically, root/circumstantial modals are usually viewed as operating over events, see e.g. Hacquard (2011). My analysis is slightly non-standard in that $\square_{ev}$ operates over what is present in the context. Evidential bias is a discourse notion, and so it is expected that it differs from other modal elements like modal verbs or adverbs.

stresses that what they require is contextual evidence (i.e., they cannot be uttered out of the blue). In fact, (14) shows that the speaker does not necessarily have to have any prior attitude to Petr’s bringing a pen to class (the speaker might be agnostic about it) in order to felicitously utter the PQ. However, the speaker might have a concrete belief about p , as in (15), where the speaker believed Petr to buy a cake as he was supposed to do it. In (18) I show that the speaker might have believed $\neg p$ (Eva not going to a concert) and ask an inner/medial negation PQ. Therefore, I assume that the interpretation of an inner nor medial negation PQ depends on speaker bias.

(18) *Scenario:* Eva is known for being an introvert and for not liking to go out in the evenings. This summer, Eva’s friends went to a lot of outdoor concerts in their hometown, although Eva stayed at home. The speaker finds out about Eva having stayed at home and asks:

- | | | |
|----|--|--------|
| a. | Eva nešla s ostatními na žádný koncert? | INNER |
| | Eva NEG.went with others to DET.NCI concert | |
| | ‘Eva didn’t go any concert with others?’ | |
| b. | Eva nešla s ostatními na nějaký koncert? | MEDIAL |
| | Eva NEG.went with others to DET.PPI concert | |
| | ‘Eva didn’t go to some concert with the others?’ | |

In Czech PQs, there is a reason to distinguish inner and medial negation based on their LF scope and the strength of evidential bias they signal. Next, I look at outer negation.

3.2 Outer negation and FALSUM focus

Let us turn to the third negation type. Semantically, I analyze this type as the Common Ground managing operator FALSUM, whose variant for Czech has been proposed by Šimík (to appear), based on Repp (2013) and Romero (2019). I give its denotation in (19).

$$(19) \quad \llbracket \text{FALSUM}_1^{\text{CZ}} \rrbracket^g(p) = \lambda w : \exists w' \in \text{EPI}_{g(1)}(w) [p(w') = 1]. p \notin \text{CG}_w$$

The content of FALSUM in (19) could be informally summarized as: the attitude holder considers it possible that the prejacent p is true, but it is not part of the CG. Its interpretation relies on the polarity of the speaker’s prior epistemic attitude: it must not be negative, i.e., the speaker must not have considered possible that $\neg p$, as shown in (20).

(20) *Scenario:* Eva is doing research for a school presentation. Yesterday she went to the school library, but didn’t borrow any book, as she worked on her laptop only. The speaker knows all this and asks:

- # Nepůjčila si tam Eva nějakou encyklopedii?
NEG.borrowed REFL there Eva DET.PPI encyclopedia
‘Didn’t Eva borrow some encyclopedia?’

The interpretation of an outer negation PQ does not depend on contextual evidence, as it is compatible with contextual evidence supporting p , against p as well as agnostic about p . This has been already observed by Staňková & Šimík (2025) and it is shown in (21), where p = ‘Petr ordered some lemonade there’. Due to this under-specification as to the value (or even the presence) of evidential bias, outer negation PQs are not expected to include the evidential modal operator introduced in Section 3.1 for medial and inner negation PQs.

(21) *Scenario*: A and B talk about their mutual friend, Petr, who went to a company dinner last night.

- A₁ Petr si prý objednal nealko. for *p*
 Petr REFL allegedly ordered non-alcoholic-drink
 ‘Allegedly, Petr ordered a non-alcoholic drink.’
- A₂ Petr si prý objednal pivo. against *p*
 Petr REFL allegedly ordered beer
 ‘Allegedly, Petr ordered a beer.’
- A₃ Petrovi se v podniku prý líbilo. agnostic about *p*
 Petr REFL in restaurant allegedly liked
 ‘Allegedly, Petr liked the restaurant.’
- B Neobjednal si tam Petr nějakou limonádu?
 NEG.ordered REFL there Petr DET.PPI lemonade
 ‘Didn’t Petr order some lemonade there?’

An important assumption of my analysis is that FALSUM is obligatorily focus marked (something similar has been proposed for high negation in English PQs by Bill & Koev (2022)). The focus targets an element encoding discourse polarity, i.e., that the proposition either *is* or *is not* in the CG. By applying focus on FALSUM, the speaker implies there is an alternative CG management strategy present in the discourse which is treating *p* as a part of the CG. More specifically, it highlights the fact that what is debated is the contrast on polarity. The focus semantic value of focused FALSUM is in (22) (in the sense of Rooth’s (1992) alternative semantics, where focus gives rise to ordinary and focus semantic values). Both FALSUM’s bias statement as well as its focus semantic value appear at the non-at-issue level of the PQ meaning.¹¹

$$(22) \quad \llbracket [\text{FALSUM}]_F \rrbracket_f(p) = \{\lambda w[p \notin \text{CG}_w], \lambda w[p \in \text{CG}_w]\}$$

From a syntactic point of view, I assume a phrase called PolP higher in the structure, which hosts $[\text{FALSUM}]_F$ and ensures its F-marking. The negative marker on the verb is related to Pol^0 via Agree; see (23), where the *Q* morpheme delivers PQ semantics.¹²

$$(23) \quad [\text{CP } Q [\text{PolP } [\text{FALSUM}]_F \text{Pol}^0 [\text{iPol}] [\text{TP SUBJECT NEG-V}_{[\text{uPol}]}]]]$$

The agreement between Pol^0 and the negative marker is indicated in two ways: either by syntactic movement of the negative verb to Pol^0 (the negative marker moves with the verb, since they are inseparable), or by shifting sentence stress onto the negative verb. In fact, these are the two ways of marking that negation is interpreted as $[\text{FALSUM}]_F$, and they lead to different word order variants. The two available word order strategies – the interrogative and the declarative one – are shown in (24) as VSO and SVO, respectively. I use CAPITAL LETTERS in the relevant parts of the examples to indicate sentence stress.¹³

¹¹I do not take polarity focus on FALSUM as an instance of VERUM focus (Höhle 1992). The former targets the CG status of the proposition, while the latter one targets its truth.

¹²In line with Groenendijk & Stokhof (1984), I assume the *Q* morpheme takes a proposition as its argument and returns a partition of the set of worlds (relative to the evaluation world *w*) such that the worlds are either compatible with *p* or with $\neg p$; its denotation is in (i). Formally, Czech PQs are signaled by a final rising tone. Since speaker bias is a non-at-issue meaning, it does not interact with *Q* in the final PQ denotation.

$$(i) \quad \llbracket [Q] \rrbracket = \lambda p_{\langle s,t \rangle} \lambda w_s \lambda q_{\langle s,t \rangle} [q = p \vee q = \neg p]$$

¹³It is possible that the fronted verb, e.g. in B₁’s question in (24), carries prenuclear prosodic prominence, but a more detailed analysis of empirical data would be needed to support this.

- (24) A₁ Eva si prý koupila motorku.
 Eva REFL allegedly bought motorbike
 ‘Allegedly, Eva has bought a motorbike.’
 B₁ Nekoupila si Eva náhodou nějaké AUTO? VSO
 NEG.bought REFL Eva NÁHODOU DET.PPI car
 ‘Has Eva bought a CAR, by any chance?’
 A₂ Petr si prý koupil auto.
 Petr REFL allegedly bought car
 ‘Allegedly, Petr has bought a car.’
 B₂ A Eva si náhodou NEKOUPILA nějaké auto? SVO
 and Eva REFL NÁHODOU NEG.bought DET.PPI car
 ‘What about Eva? Hasn’t she bought some car, by any chance?’

The interrogative word order in ((24)-B₁) is the result of the negative verb having moved to Pol⁰; this is schematically illustrated in (25). This is assumed to be the default way of signaling outer negation.

- (25) [CP Q [PolP [FALSUM]_F NEG-V [TP SUBJECT t_V OBJECT]]]

The fact that polarity focus leads to polarity fronting has been observed also for other languages, e.g. by Giurgea & Remberger (2014) for Sardinian and Romanian or by Holmberg (2013) for English. It seems that the verb fronting induced by focus on FALSUM (as well as verb fronting in positive unbiased Czech PQs) could mirror the movement of wh-words in wh-questions (see Truckenbrodt (2012) on focus marking of wh-words in English and their syntactic behavior). In Czech, both of these involve movement of material to the left periphery with the intention to mark focus as well as clause type (in the sense of Cheng (1997)). I leave the investigation of this parallel of focus fronting across different types of questions for future research.

The declarative variant in ((24)-B₂) is motivated by the presence of a contrastive topic (= CT), which tends to occupy the initial position in Czech utterances (Sturgeon 2006; Constant 2014). If a CT is present in the structure, the negative verb stays in situ, although focus on FALSUM is still interpreted, see (26). This is ensured by placing sentence stress on the negative verb, which indicates agreement with Pol⁰.¹⁴ Hypothetically, the CT-NEG V word order is driven by the semantics of these two elements: a CT needs to outscope a focus, which overtly leads to the CT-NEG V sequence in Czech (see Wagner (2012) on the interaction of CT with focus in German). This is a possible reason for the verb staying in situ in declarative outer negation PQs, but it is not in the scope of this article to discuss this more general issue.

- (26) [CP Q [PolP [FALSUM]_F Pol⁰_[iPol] [TP [SUBJECT]_{CT} NEG-V_[uPol]]]]

When it comes to the opposing tendencies of CT- vs. verb-fronting, it seems that the former one is stronger than the latter. By this I mean that the valuation of [uPol] by syntactic movement, which is probably the default strategy, can be overridden by CT movement to the left

¹⁴An anonymous reviewer points out that my analysis bears resemblance to the account of Romero & Han (2004) in pursuing a parallel analysis of high negation and in-situ verum accent. However, my account draws a line between outer negation on one side and medial and inner negations on the other in that outer negation (FALSUM) is obligatorily focused (no matter if syntactically high or low), hence the obligatory presence of Pol⁰. The other two negations might, but need not be focused. Importantly, I take focus on FALSUM and focus on medial/inner negation to be of a different sort (more on that in Section 4). In this way, my account differs even from Goodhue (2022).

periphery. The fact that outer negation PQs carry *FALSUM* focus has further implications for information structure as well as for the formal encoding of PQs.

4 Focus encoding in negative PQs

Depending on which negation reading is activated, PolP either is or is not present. As we have seen in Section 3.2, only outer negation (*FALSUM*) needs to be focused, since PolP is present, while the two structurally lower negations do not. More precisely, medial and inner negation PQs might encode focus on polarity (although this is only a possibility), but this seems to be qualitatively different from the focus on *FALSUM*, while polarity focus in medial and inner negation PQs requires all the lexical material in the clause to be contextually given. Given elements require a synonymic or a hyponymic antecedent in the immediately preceding discourse (Rochemont 2016). In Czech, givenness of an element leads to its deaccentuation (Šimík & Wierzbica 2015, 2017; Groeben et al. 2017; Hamlaoui et al. 2018). In ((27)-B₁) the word *knížku* ‘book’ counts as contextually given, since it has a synonymic (identical) antecedent in A’s utterance. The word *encyklopedii* ‘encyclopedia’ in ((27)-B₂) does not count as given, since it lacks a synonymic or hyponymic antecedent (*knížku* ‘book’ in A’s utterance is a hyperonym).

- (27) A Eva si prý v knihovně nepůjčila knížku.
 Eva REFL allegedly in library NEG.borrowed book
 ‘Allegedly, Eva didn’t borrow a book in the library.’
- B₁ Eva si fakt NEPŮJČILA {žádnou / nějakou} knížku?
 Eva REFL FAKT NEG.borrowed DET.NCI DET.PPI book
 ‘Eva didn’t borrow any / some book?’
- B₂ #Eva si fakt NEPŮJČILA {žádnou / nějakou} encyklopedii?
 Eva REFL FAKT NEG.borrowed DET.NCI DET.PPI encyclopedia
 Intended: ‘Eva didn’t borrow any / some encyclopedia?’

In this, polarity focus on medial/inner negation patterns with polarity focus in positive PQs; see (28) where *auto* ‘car’ in B₁ is given, while *škodovku* ‘Skoda car’ in B₂ is not.

- (28) A Eva si prý koupila auto.
 Eva REFL allegedly bought car
 ‘Allegedly, Eva bought a car.’
- B₁ Eva si KOUPILA nějaké auto?
 Eva REFL bought DET.PPI car
 ‘Eva DID buy some car?’
- B₂ #Eva si KOUPILA nějakou škodovku?
 Eva REFL bought DET.PPI Skoda-car
 Intended: ‘Eva DID buy some Skoda car?’

FALSUM focus, on the contrary, requires at least one lexical element to be contextually new; see (29) with the interrogative outer negation PQs.¹⁵ The word *román* ‘novel’ in B₂ is new, not given, since *knížku* ‘book’ in A’s utterance is its hyperonym, not hyponym.

¹⁵Note that the same is true for declarative outer negation PQs, where the contextually new element is the contrastive topic, as in ((24)-B₂).

- (29) A Eva si prý v knihovně nepůjčila knížku.
 Eva REFL allegedly in library NEG.borrowed book
 ‘Allegedly, Eva didn’t borrow a book in the library’
- B₁ #Nepůjčila si tam Eva náhodou nějakou KNÍŽKU?
 NEG.borrowed REFL there Eva NÁHODOU DET.PPI book
 Intended: ‘Didn’t Eva borrow some BOOK there, by any chance?’
- B₂ Nepůjčila si tam Eva náhodou nějaký ROMÁN?
 NEG.borrowed REFL there Eva NÁHODOU DET.PPI novel
 ‘Didn’t Eva borrow some NOVEL there, by any chance?’

In general, outer negation PQs seem to be restricted in encoding information structure, which happens because of the presence of Pol⁰. Medial and inner negation PQs do not feature Pol⁰, which enables focusing of the lexical content of some constituent, e.g. a VP or a verb; see the examples in (30).

- (30) A₁ Allegedly, Eva didn’t enjoy the birthday party last night.
- B₁ Eva tam ne[hrála nějaké HRY]_F? MEDIAL, VP foc
 Eva there NEG.played DET.PPI games
 ‘Eva didn’t play some GAMES there?’
 $[[Eva [played\ games]_F]_f = \{[[Eva\ played\ games]]_o, [[Eva\ danced]]_o, [[Eva\ sang\ karaoke]]_o, \dots\}$
- A₂ Eva borrowed a lot of books from me yesterday.
- B₂ Eva si žádnou knížku [NEKOUPILA]_F? INNER, V foc
 Eva REFL DET.NCI book NEG.bought
 ‘Eva didn’t BUY any book?’
 $[[Eva\ didn’t\ [buy]_F\ a\ book]]_f = \{[[Eva\ didn’t\ buy\ a\ book]]_o, [[Eva\ didn’t\ borrow\ a\ book]]_o, \dots\}$

When focus on FALSUM is encoded by verb fronting, a constituent other than the negation itself can be F-marked to indicate (contrasting) alternatives; see (31) with focused object and verb. The F-marking is signaled by sentence stress on the focused constituent. It should further be noted that what is encoded by sentence stress in ((30)-B₂) and ((31)-B₂) is focus on the predicate (the lexical content of the verbs ‘buy’ and ‘pepper’, respectively), not on polarity, unlike in ((24)-B₂). In structures of this kind, my account predicts that two foci are present: the obligatory one on FALSUM encoded by verb fronting, and the other one on another constituent encoded by prosodic prominence.

- (31) A Eva is helping out in the kitchen with lunch preparation. So far, she salted the duck, I think.
- B₁ Neosolila Eva náhodou [KUŘE]_F? FALSUM & O foc
 NEG.salted Eva NÁHODOU chicken
 ‘Hasn’t Eva salted CHICKEN?’
 $[[Eva\ salted\ [chicken]_F]_f = \{[[Eva\ salted\ chicken]]_o, [[Eva\ salted\ duck]]_o, \dots\}$
- B₂ [NEOPEPŘILA]_F Eva náhodou tu kachnu? FALSUM & V foc
 NEG.peppered Eva NÁHODOU DET duck
 ‘Hasn’t Eva PEPPERED the duck?’
 $[[Eva\ [peppered]_F\ the\ duck]]_f = \{[[Eva\ peppered\ the\ duck]]_o, [[Eva\ salted\ the\ duck]]_o, \dots\}$

When focus on *FALSUM* is encoded by prosodic prominence, i.e. the negative verb stays in situ, F-marking on another constituent is impossible. The example in (32) shows that all constituents other than the negation and the contrastive topic need to be backgrounded for the utterance to be felicitous. This condition is met in ((32)-B₁) where ‘buy’ and ‘car’ are backgrounded (as well as contextually given). The utterance in ((32)-B₂) contrasts a car with a motorbike, and ((32)-B₃) contrasts buying with renting. Both of these latter two are predicted to be infelicitous, as sentence stress must fall on the negative verb to indicate *FALSUM* focus – it cannot be used to indicate focus on anything else.

- (32) A Eva bought a new car.
- B₁ A [Petr]_{CT} si náhodou [NE]_FKOUPIL auto? FALSUM foc
 and Petr REFL NÁHODOU NEG.bought car
 ‘And what about Petr, didn’t also he buy a car?’
- B₂ #A [Petr]_{CT} si náhodou nekoupil [MOTOR_{KU}]_F? O foc
 and Petr REFL NÁHODOU NEG.bought motorbike
 Intended: ‘And what about Petr, didn’t he buy a motorbike?’
- B₃ #A [Petr]_{CT} si náhodou NE[PŮJČIL]_F auto? V foc
 and Petr REFL NÁHODOU NEG.rented car
 Intended: ‘And what about Petr, didn’t he rent a car?’

5 Conclusion

In this article, I proposed that Czech distinguishes three types of negation: outer, medial and inner. The types are distinct in their reading, which I captured as differences in their scope on the LF structure. The three interpretations vary in the interaction with polarity items, particles and information structure and also in their distribution across clause types.

Inner negation is the standard sentential negation which operates inside of the proposition, has the narrowest semantic scope, and thus licenses NCIs. It licenses the aspectual particle *ještě* and in a PQ (in combination with the proposed modal operator) it derives strong negative evidential bias. It occurs in assertions and questions. Medial negation stays overtly in situ, but semantically, it has a wider scope than inner negation. It can scope over PPIs and it derives weak negative evidential bias in PQs. It is felicitous in polar questions, but not in assertions. Both inner and medial negation readings are compatible with the particle *fakt*. Outer negation, which I analyzed as the operator *FALSUM*, is obligatorily focused. The focus is realized either by overt movement of the verb carrying the negation (akin to wh-movement in wh-questions) – correlating with the so-called high negation – or by prosodic means in case the initial position is used for a contrastive topic. It encodes weak speaker bias to the positive prejacent of the question and licenses the particle *náhodou*. It occurs specifically in polar questions.

My account has further implications for information structure, esp. further focusing with the intention to encode a contrast, in the scope of negation. While a declarative polar question with outer negation cannot contain another focused constituent than the negation itself, focusing in other questions is not limited and happens freely.

The material of polar questions proved to be suitable to show that negation in Czech is a complex interface phenomenon. To capture and distinguish its three interpretations I examined their syntactic, semantic as well as pragmatic characteristics.

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